

Does preserving species fraction preserve measured coarsening?

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The question

After reducing image resolution, is matching the original species fraction enough to preserve a fitted finite-window coarsening exponent? The comparison is with the **original image measurement**, not with a forced value of $1/3$. Lower early slopes in Kawasaki simulations motivated this question; this experiment does not settle every cause of that original shortfall.



Figure 1. The same 50:50 snapshot, not three simulations. A fixed example uses the first trajectory at 1,119 sweeps and tie priority 110. White-site fraction is 0.5 in both binary images; averaging retains grey values. The example illustrates the operation, not the statistical result.

The controlled test

Sixteen fresh independent trajectories (eight per mixture) used a 128 x 128 periodic lattice, Kawasaki exchange with Metropolis acceptance and $J_x = J_y = 1$. After 200 preparation sweeps at $3 T_c$, each was quenched to $0.65 T_c$. A protocol and source hashes were fixed before these runs, after an earlier processing effect was known. The primary comparison uses 24 common checkpoints from 1,119 to 20,000 post-quench sweeps.

Mixture	Original exponent	Matched exponent	Difference [95% interval]
50:50	0.23141	0.18094	-0.05047 [-0.05293, -0.04782]
15:85	0.24214	0.16563	-0.07651 [-0.08006, -0.07255]

Main finding: at fourfold reduction, fraction matching did not preserve the original fitted exponent. Matching is exact at 50:50; the 15:85 coarse fraction is rounded to the nearest attainable pixel count. Finer reduction can give much smaller differences: this is not a claim that matching never helps.

Intervals use 1,000 paired whole-trajectory bootstrap draws, not independent checkpoints. They quantify run-sampling uncertainty conditional on this analysis, not uncertainty in real-alloy physics.

Why use this short window when longer runs exist?

A separate 128-trajectory campaign reached one million sweeps across four lattice sizes and two mixtures. At $L = 128$ its periodic half-height fits over 21,336-1,000,000 sweeps were 0.300 (50:50) and 0.281 (15:85). Another 40-run, $0.6 T_c$ reference campaign reached 4.5 million sweeps; its literature-matched analysis is unfinished. Neither establishes an asymptotic plateau. The short test above isolates observation changes on identical snapshots in a deliberately demanding resolution regime; it is not the entire simulation study.

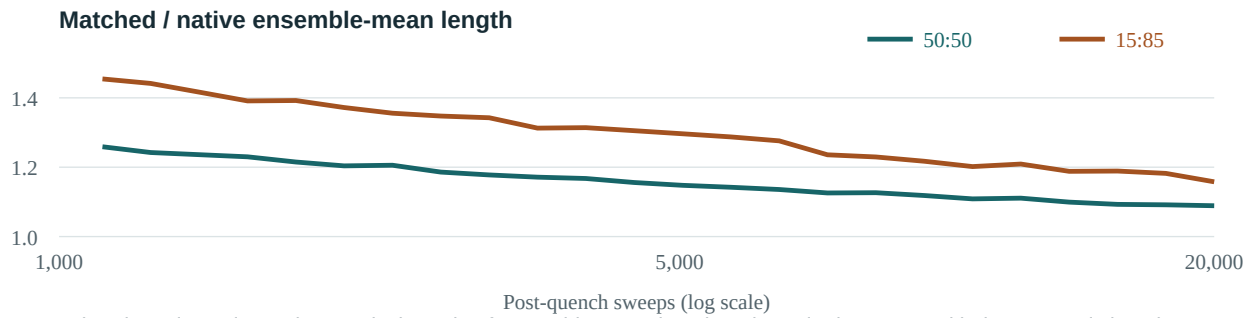


Figure 2. Fresh trajectories only; each curve is the ratio of ensemble-mean lengths. The ratio decreases with time, so early lengths are inflated more than late ones and the fitted slope is reduced. Constant multiplicative length error would leave the slope unchanged. This accounting identity does not identify a unique geometric cause.

How the measurement works

Lengths are the mean X/Y half-height crossings of variance-normalized, mean-subtracted covariance, using valid pairs without wrapping image edges. Coarse lengths are converted to original lattice spacing. Log ensemble-mean length is fitted against log sweeps on shared times. Rank segmentation retains the highest block means; five fixed tie priorities are averaged within each trajectory. These are not additional independent runs.

What the result does not establish

Some coarse lengths approach or fall below one pixel. Four grid/window origins and a second covariance length retained the negative shift, but an exploratory chord analysis found smaller shifts. No universal sensitivity is claimed. Two proposed exponent-warning screens accepted zero of 18 comparisons: they are not validated selectors. A direct-pair audit corrected a roundoff issue in factor-eight secondary results; the factor-four result was unchanged.

A separate materials-image audit

The audit compares calibrated binary masks at native, 2x and 4x spacing and reports length, phase fraction and resolution warnings. Across all 81 FeM iron-ore fields [5], median length changes were -0.898% and +1.610%. A post-hoc opposite tie rule changed the 2x median to +1.925%. All fields were resolved and unflagged, unlike the short-window exponent screens. These are fields from one specimen, not 81 specimens; producer masks are references, not physical truth. This is a static sensitivity test, not ageing validation, microscope calibration or laboratory adoption. Phase fraction is not generally chemical composition.

Closest work and the possible contribution

Zabler et al. [1] already tested coarsening fits against segmentation threshold; Ledesma-Alonso et al. [2] studied resolution-dependent descriptors. Majumder and Das [3,4] used noise filtering and initial-length/finite-size analysis; their three length measures agreed up to constant factors. Our short raw fits are not equivalent. The possible contribution is the paired dynamic benchmark after fraction matching, not discovery of image bias or a new growth law. Novelty remains unestablished.

Three questions for a reviewer

1. Does this specific benchmark add anything useful beyond the closest prior work?
2. Are the length definition, common time mask and paired uncertainty defensible?
3. Could a static mask comparison inform an imaging decision in your work, and what reference and tolerance would make a fair pilot?

[1] Zabler et al., *Acta Materialia* 55 (2007), 5045-5055. doi:10.1016/j.actamat.2007.05.028

[2] Ledesma-Alonso et al., *Physical Review E* 97 (2018), 023304. arXiv:1712.03183

[3] Majumder and Das, *Physical Review E* 81 (2010), 050102(R). arXiv:1001.3985

[4] Majumder and Das (2013), arXiv:1305.2556. [5] Gomes et al., FeM dataset, doi:10.5281/zenodo.5014700.

Code, design, analysis and this draft used substantial AI assistance. Student approval and an accessible evidence copy must be checked before sharing. No independent review or outside use has occurred. This requests criticism, not endorsement.